

CS257 Linear and Convex Optimization

Homework 10

Due: November 30, 2020

November 23, 2020

1. Consider the equality constrained least squares problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \|\mathbf{Ax} - \mathbf{b}\|^2 \\ \text{s.t.} \quad & \mathbf{Gx} = \mathbf{h} \end{aligned}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $\text{rank } \mathbf{A} = n$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{G} \in \mathbb{R}^{k \times n}$ with $\text{rank } \mathbf{G} = k$.

- (a). Find KKT system of this problem
- (b). Find a closed form solution for the optimal solution $\mathbf{x}^*$ and the corresponding Lagrange multiplier $\boldsymbol{\lambda}^*$.
- (c). Find the KKT system and $\mathbf{x}^*$ and $\boldsymbol{\lambda}^*$ for

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 10 \\ -5 \end{bmatrix}, \quad \mathbf{G} = (1, 1), \quad \mathbf{h} = 1$$

- (d). Find the optimal solution $\mathbf{x}^*$ for the $\mathbf{A}, \mathbf{b}, \mathbf{G}, \mathbf{h}$ in (c) by reduction to an unconstrained problem.

Solution. (a). The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \|\mathbf{Ax} - \mathbf{b}\|^2 + \boldsymbol{\lambda}^T (\mathbf{Gx} - \mathbf{h})$$

The Lagrange condition is

$$\begin{cases} \nabla_{\mathbf{x}} \mathcal{L} = 2\mathbf{A}^T (\mathbf{Ax} - \mathbf{b}) + \mathbf{G}^T \boldsymbol{\lambda} = \mathbf{0} \\ \nabla_{\boldsymbol{\lambda}} \mathcal{L} = \mathbf{Gx} - \mathbf{h} = \mathbf{0} \end{cases} \quad (\text{S1})$$

The KKT system is

$$\begin{bmatrix} 2\mathbf{A}^T \mathbf{A} & \mathbf{G}^T \\ \mathbf{G} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} 2\mathbf{A}^T \mathbf{b} \\ \mathbf{h} \end{bmatrix}$$

- (b). Since $\mathbf{A}$ has full column rank, $\mathbf{A}^T \mathbf{A} \succ \mathbf{O}$ and hence invertible. From the first equation of (S1),

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} (\mathbf{A}^T \mathbf{b} - \frac{1}{2} \mathbf{G}^T \boldsymbol{\lambda}) \quad (\text{S2})$$

Plugging into the second equation,

$$\mathbf{G}\mathbf{x} = \mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1}(\mathbf{A}^T \mathbf{b} - \frac{1}{2} \mathbf{G}^T \lambda) = \mathbf{h} \implies \lambda^* = 2[\mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{G}^T]^{-1}[\mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} - \mathbf{h}]$$

where we have used the fact $\mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{G}^T \succ \mathbf{O}$ and hence invertible. To see that $\mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{G}^T \succ \mathbf{O}$, first note that $(\mathbf{A}^T \mathbf{A})^{-1} \succ \mathbf{O}$, since $\mathbf{A}^T \mathbf{A} \succ \mathbf{O}$. Since $\mathbf{G}^T$ has full column rank, $\mathbf{y} \triangleq \mathbf{G}^T \mathbf{x} \neq \mathbf{0}$ for $\mathbf{x} \neq \mathbf{0}$. Thus for $\mathbf{x} \neq \mathbf{0}$,

$$\mathbf{x}^T \mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{G}^T \mathbf{x} = \mathbf{y}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{y} > 0$$

Plugging the expression for λ^* into (S2),

$$\mathbf{x}^* = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} - (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{G}^T [\mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{G}^T]^{-1} [\mathbf{G}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} - \mathbf{h}]$$

(c). The KKT system is

$$\begin{bmatrix} 12 & 2 & 1 \\ 2 & 28 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \lambda \end{bmatrix} = \begin{bmatrix} 30 \\ -70 \\ 1 \end{bmatrix}$$

The optimal solution is

$$\mathbf{x}^* = (3.5, -2.5)^T,$$

and the Lagrange multiplier is

$$\lambda^* = -7$$

(d). Rewriting the constraint as $x_2 = 1 - x_1$.

$$\mathbf{A}\mathbf{x} - \mathbf{b} = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ 1 - x_1 \end{bmatrix} - \begin{bmatrix} 5 \\ 10 \\ -5 \end{bmatrix} = \begin{bmatrix} x_1 - 4 \\ 4x_1 - 13 \\ -x_1 + 7 \end{bmatrix}$$

Thus

$$g(x_1) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = (x_1 - 4)^2 + (4x_1 - 13)^2 + (x_1 - 7)^2$$

Minimizing g ,

$$g'(x_1) = 0 \implies x_1^* = 3.5$$

and

$$x_2^* = 1 - x_1^* = -2.5$$

□

2. We considered the following convex optimization problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \end{aligned} \tag{1}$$

Suppose $\mathbf{Q} \succeq \mathbf{O}$. The problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & h(\mathbf{x}) \triangleq f(\mathbf{x}) + (\mathbf{Ax} - \mathbf{b})^T \mathbf{Q} (\mathbf{Ax} - \mathbf{b}) \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \end{aligned} \tag{2}$$

is equivalent to the problem in (1).

- (a). Suppose we want to use Newton's method to solve (2). Find the KKT system used to solve for the Newton direction at a feasible $\mathbf{x}_0$.
- (b). If we use Newton's method to solve both (1) and (2), are the Newton directions the same at the same feasible point $\mathbf{x}$? Are the corresponding Lagrange multipliers (the $\boldsymbol{\lambda}$ in the KKT system) the same?

Solution. (a). The gradient and the Hessian of h at $\mathbf{x}_0$ are

$$\begin{aligned} \nabla h(\mathbf{x}_0) &= \nabla f(\mathbf{x}_0) + 2\mathbf{A}^T \mathbf{Q} (\mathbf{Ax}_0 - \mathbf{b}) = \nabla f(\mathbf{x}_0) \\ \nabla^2 h(\mathbf{x}_0) &= \nabla^2 f(\mathbf{x}_0) + 2\mathbf{A}^T \mathbf{Q} \mathbf{A} \end{aligned}$$

The KKT system is

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}_0) + 2\mathbf{A}^T \mathbf{Q} \mathbf{A} & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{d} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}_0) \\ \mathbf{0} \end{bmatrix}$$

- (b). Multiplying the second equation by $-2\mathbf{A}^T \mathbf{Q}$ and adding the result to the first equation,

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}_0) & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{d} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}_0) \\ \mathbf{0} \end{bmatrix}$$

which is the same KKT system as that for (1), so both the Newton directions and the corresponding Lagrange multipliers are the same for (1) and (2). □

3. Let $\mathbf{x} \in \mathbb{R}^3$. Consider

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = e^{x_1} + e^{2x_2} + e^{2x_3} \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1 \end{aligned} \tag{3}$$

- (a). Solve problem (3) by the Lagrange multiplier method. Show the optimal solution $\mathbf{x}^*$, the Lagrange multiplier λ^* and the optimal value f^* .
- (b). Find the closed-form expression for the Newton direction at a feasible $\mathbf{x}$ by solving the KKT system. Write a simple loop to run 5 iterations of Newton's method with the direction you find, initial point $\mathbf{x}_0 = (1, 0, 0)^T$, and constant step size $t = 1$. Show $\mathbf{x}$ for each iteration.
- (c). Implement the algorithm on slide 24 of Lecture 11 in the `newton_eq` function of `newton.py`. Then use your implementation to solve (3) with initial points $\mathbf{x}_0 = (1, 0, 0)^T$ and $\mathbf{x}_0 = (2, -1, 0)^T$. Show the outputs.

Solution. (a). The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \lambda) = e^{x_1} + e^{2x_2} + e^{2x_3} + \lambda(x_1 + x_2 + x_3 - 1)$$

The optimality condition is

$$\begin{cases} e^{x_1} + \lambda = 0 \\ 2e^{2x_2} + \lambda = 0 \\ 2e^{3x_3} + \lambda = 0 \\ x_1 + x_2 + x_3 - 1 = 0 \end{cases}$$

Thus

$$\begin{cases} x_1 = \log(-\lambda) \\ x_2 = \frac{1}{2} \log(-\lambda) - \frac{1}{2} \log 2 \\ x_3 = \frac{1}{2} \log(-\lambda) - \frac{1}{2} \log 2 \end{cases}$$

Substitution into the constraint yields,

$$2 \log(-\lambda) = 1 + \log 2 \implies \lambda^* = -\sqrt{2e}$$

$$\begin{cases} x_1^* = \frac{1+\log 2}{2} \\ x_2^* = \frac{1-\log 2}{4} \\ x_3^* = \frac{1-\log 2}{4} \end{cases}$$

and

$$f^* = -2\lambda^* = 2\sqrt{2e}$$

(b). The KKT system is

$$\begin{bmatrix} e^{x_1} & 0 & 0 & 1 \\ 0 & 4e^{2x_2} & 0 & 1 \\ 0 & 0 & 4e^{2x_3} & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \lambda \end{bmatrix} = \begin{bmatrix} -e^{x_1} \\ -2e^{2x_2} \\ -2e^{2x_3} \\ 0 \end{bmatrix}$$

$$\begin{aligned} d_1 &= -1 + \frac{2e^{-x_1}}{e^{-x_1} + \frac{1}{4}e^{-2x_2} + \frac{1}{4}e^{-2x_3}} \\ d_2 &= -\frac{1}{2} + \frac{\frac{1}{2}e^{-2x_2}}{e^{-x_1} + \frac{1}{4}e^{-2x_2} + \frac{1}{4}e^{-2x_3}} \\ d_3 &= -\frac{1}{2} + \frac{\frac{1}{2}e^{-2x_3}}{e^{-x_1} + \frac{1}{4}e^{-2x_2} + \frac{1}{4}e^{-2x_3}} \\ \lambda &= \frac{-2}{e^{-x_1} + \frac{1}{4}e^{-2x_2} + \frac{1}{4}e^{-2x_3}} \end{aligned}$$

The output is

```
iteration 0 : [1. 0. 0.]
iteration 1 : [0.84776623 0.07611688 0.07611688]
iteration 2 : [0.84657359 0.0767132 0.0767132 ]
iteration 3 : [0.84657359 0.0767132 0.0767132 ]
iteration 4 : [0.84657359 0.0767132 0.0767132 ]
iteration 5 : [0.84657359 0.0767132 0.0767132 ]
```

(c). For $\mathbf{x}_0 = (1, 0, 0)^T$,

```
iteration 0: [1. 0. 0.]
iteration 1: [0.84776623 0.07611688 0.07611688]
iteration 2: [0.84657359 0.0767132 0.0767132 ]
iteration 3: [0.84657359 0.0767132 0.0767132 ]
optimal value: 4.663287963194248
step sizes: [1.0, 1.0, 1.0]
```

For $\mathbf{x}_0 = (2, -1, 0)^T$,

```
iteration 0: [ 2. -1.  0.]
iteration 1: [ 1.12123562  0.15481017 -0.2760458 ]
iteration 2: [ 0.9666166  0.09922011 -0.0658367 ]
iteration 3: [0.90354278 0.08470489 0.01175234]
iteration 4: [0.87442649 0.079997  0.0455765 ]
iteration 5: [0.86035569 0.07818861 0.0614557 ]
iteration 6: [0.85343009 0.07741064 0.06915928]
iteration 7: [0.84999338 0.07705202 0.0729546 ]
iteration 8: [0.84828139 0.07688016 0.07483845]
iteration 9: [0.84742697 0.07679607 0.07577696]
iteration 10: [0.84700015 0.07675448 0.07624536]
iteration 11: [0.84678684 0.07673381 0.07647936]
iteration 12: [0.84668021 0.0767235 0.0765963 ]
iteration 13: [0.8466269 0.07671835 0.07665476]
iteration 14: [0.84660024 0.07671578 0.07668398]
iteration 15: [0.84658692 0.07671449 0.07669859]
iteration 16: [0.84658025 0.07671385 0.0767059 ]
iteration 17: [0.84657692 0.07671353 0.07670955]
optimal value: 4.663287963238539
step sizes: [1.0, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5,
0.5, 0.5, 0.5]
```

A possible implementation

```
def newton_eq(f, fp, fpp, x0, A, b, initial_stepsize=1.0,
             alpha=0.5, beta=0.5, tol=1e-5, maxiter=100000):
    x_traces = [np.array(x0)]
    stepsize_traces = []

    x = np.array(x0)
    m = len(b)
    for it in range(maxiter):
        grad = fp(x)
        H = fpp(x)
```

```

# Newton direction
K = np.block([[H, A.T], [A, np.zeros((m,m))]])
x = np.linalg.inv(K)@np.block([-grad, np.zeros(m)])
dx = dx[0:len(x0)]

# line search
fx = f(x)
alpha_grad_dx = alpha * (grad@dx)

stepsize = initial_stepsize
while f(x + stepsize * dx) > fx + stepsize * alpha_grad_dx:
    stepsize *= beta

x += stepsize * dx
x_traces.append(np.array(x))
stepsize_traces.append(stepsize)

if np.linalg.norm(dx) < tol:
    break

return x_traces, stepsize_traces

```

□